



Motion Attitude and Aerodynamic Characteristics Research of Flapping Wings Driven by Micro Servoactuator

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Abstract

Compared to the traditional flapping-wing structure with single motion mode, a micro servoactuator driven Flapping-Wing Air Vehicle (FWAV) breaks free from the limitations imposed by the motion parameters of the crank-connecting rod mechanism. It allows for simultaneous control of wings' position and velocity attitude through pulse width modulation, showcasing unrivaled controllability and promising extensive applications. However, this method of motion control also brings new challenges to the design of the wings' motion parameters. This study seeks to investigate the relationship between the motion parameters of micro servoactuator driven FWAV and its aerodynamic characteristics, then explore a servo control method that can optimize its thrust-producing performance. To achieve this, this paper involves the establishment of Amplitude Loss Model (ALM), Flapping Wing Dynamic Model (FWDm), and Power Load Model (PLM), followed by motion capture experiments, dynamic monitoring experiments, and power monitoring experiments. Experimental results show that the proposed modeling method, which fully considers the amplitude loss effect and advanced twisting effect in flapping-wing motion, can accurately calculate thrust, power, and power load, with prediction errors of less than 10%, 5% and 13%, respectively. This high-precision model can effectively optimize motion parameters, allowing for better performance of flapping-wing motion.

Keywords Flapping wing · Micro servoactuator · Motion attitude · Aerodynamic model

1 Introduction

In recent years, the rapid development of micro-electromechanical technology has enabled human beings to actualize the flapping flight that has been imagined for thousands of years on the scale of micro drones [1, 2]. However, there are still numerous formidable technical challenges that impede progress in the field of flapping flight. The aerodynamic force generated by flapping-motion cannot be comprehensively explained [3], and there is a dearth of research on its motion characteristics. In contrast to conventional fixed-wing or rotary-wing aerial vehicles, FWAVs lack external power sources, such as propellers and jet engines. Its

driving force is exclusively generated by the wings flapping the air under low Reynolds number conditions [4], exhibiting pronounced time-varying, nonlinear characteristics that are challenging to quantify [5]. Simultaneously, it exhibits a strong dependence [6] on factors such as the flapping amplitude, frequency, wing surface shape, Control Signal Shape Function (CSSF), and etc.

The flight mechanisms of flying creatures or FWAVs at different scales exhibit significant divergence [7]. For instance, the clap-and-fling mechanism [8] has been demonstrated as an unsteady mechanism that effectively enhances the aerodynamic forces generated during flapping motion. This mechanism is commonly employed in quad-wing FWAVs [9–11], or in cases where the overall mass of the aircraft is around 20 g, with the flapping amplitude often approaches or exceeds 180 degrees [12, 13]. Nonetheless, due to mechanical constraints, maintaining high-frequency and large-amplitude oscillations proves challenging, making it nearly impractical for bird-like FWAVs to generate aerodynamic advantages. Though bird-like FWAVs exhibit a lower flapping frequency, the mechanism of generating

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aerodynamic force remains equally complex. The intricate interplay between flexible wing deformation under various motion postures and the upstream flow conditions has become a significant challenge in its dynamic analysis [3, 14].

At the current stage, bird-like FWAVs hold greater practical value due to its balance between load-bearing capacity and concealment. Generally, bird-like FWAVs are designed as typical underactuated system [15, 16] due to the absence of flexible wing deformation control. This underactuated system has only one degree of freedom dedicated to flapping-wing motion, typically achieved using crank-rocker mechanisms such as planar asymmetric single-crank single-rocker [17], planar symmetric double-crank double-rocker [18] and spatial symmetric double-crank double-rocker [16]. Their output motion modes closely resemble simple harmonic rotational motion in the plane. The advantage of this drive system lies in its ability to utilize motors with superior power density characteristics, thereby achieving more robust propulsion performance. However, its limitations are equally conspicuous. Once the structural length parameters are established, they cannot be changed. The wings can only execute periodic motions according to a predetermined set of motion parameters. The control system's adjustment range is typically confined to regulating the flapping frequency, while other factors such as the flapping stroke angle and wing trajectory are not within the constraints of the control system. On the one hand, this presents a significant hurdle in the investigation of flapping motion postures. The relationship between flapping frequency and amplitude is often arbitrarily configured. On the other hand, it diminishes the controllability of FWAV, hampering their ability to fully exploit their maneuverability advantages.

The RoboRaven series prototype [19–23] may be the first FWAV to embrace the micro servoactuator driven scheme. Diverging from traditional flapping-wing motion generation mechanisms, the micro servoactuator driven FWAV obviates the need for intricate mechanical structure design. It can rapidly achieve simultaneous control of wing position and speed through pulse width modulation. The amplitude control, phase control, and even motion trajectory control that cannot be realized by the traditional flapping-wing motion generation mechanism can be realized by the micro

servoactuator driven scheme. The high controllability of this emerging actuation scheme endows it with significant development potential, but it also brings new challenges to the design of flapping motion attitude. The speed performance limitations of the driving mechanism cause amplitude attenuation during frequent forward and reverse rotations [22, 23], leading to a corresponding reduction in the aerodynamic performance of FWAVs. This phenomenon is unavoidable but is expected to minimize its negative impact by optimizing the motion attitude. Huang et al. [24, 25] were the first to consider the impact of different CSSFs on wing motion waveforms and aerodynamic characteristics. They found that the flapping gait with a square wave signal could achieve the maximum motion stroke at frequencies below 3 Hz. Although they did not provide a detailed explanation for this phenomenon, it suggests an effective method for reducing speed loss. Additionally, since servo-driven FWAVs do not have fixed parameter limitations, the flapping amplitude can be variable. Different parameter configurations can result in significant changes in aerodynamic characteristics. Research on how to reasonably configure amplitude and frequency parameters to optimize the aerodynamic performance of FWAVs is still lacking.

Therefore, this study delves into the motion attitude and dynamic performance of the micro servoactuator driven FWAV, employing both theoretical analysis and experimental methods. The rest of the article is organized as follows. Section 2 provides the detailed implementation approach of the micro servoactuator driven scheme in the electromechanical system. Section 3 establishes the ALM, FWDM, and PLM for the servoactuator driven flapping wing motion to quantify the negative impact of the amplitude loss phenomenon. Section 4 experimentally validates the proposed theory using high-speed cameras and six-dimensional force sensors and introduces the method to improve the aerodynamic characteristics of FWAVs by optimizing the motion attitude. Section 5 summarizes the results and outlines future work.

2 Generating Flapping Wing Motion

As a common actuator, the micro servoactuator exhibits excellent closed-loop position control accuracy and has been utilized in various aircraft models, vehicle models, and desktop-level robots for an extended period. With the continuous advancement of technology, the response speed of micro servoactuator has been further enhanced, enabling it to generate flapping-wing motion. In this study, GDW's DS290MG was employed, and its key parameters are presented in Table 1. To achieve the highest response speed, the test voltage was set to the maximum value within the

Table 1 Key parameters of DS290MG

Parameter	Value
Mass	20 ± 1 g
Working frequency	1520us/333Hz
Operation range	180°
Test voltage	8.4 V
Speed	0.06 s/60°
Torque	6.0 kg·cm
Stall current	2000 mA

product's operating voltage range. Prolonged operation at 8.4 V voltage while executing maneuvers with flapping-wing motion did not result in noticeable heating of the micro servoactuator.

The micro servoactuator can generate periodic reciprocating flapping-wing motion through pulse width modulation. The working frequency of DS290MG is 333 Hz, which means the minimum identifiable electrical signal pulse width is 3ms. When the high-level time and low-level time are both set at 1.5ms, the steering gear arm remains in a neutral position. By adjusting the high-level time from 0.5ms to 2.5ms, the steering gear arm undergoes an angular displacement change of 180 degrees across its entire stroke. Through programming control of the position node and the delay function, precise control of the position and velocity within the permissible performance range is achievable.

Figure 1 illustrates the relationship between the electrical signal's pulse width and the real-time angular displacement of the steering gear arm. A sine wave CSSF is used as an example to provide a pulse width modulation method within a flapping period. In Fig. 2, t_n represents the total duration of the electrical signal at each position node, and n signifies the number of preset nodes in a flapping period, satisfying the relationship:

$$nt_n = T \quad (1)$$

where T represents the complete duration of a flapping period. When the value of n is too small, it can result in a challenge of low real-time position control accuracy and potential stalling of the flapping-wing motion. Conversely, the value of n should not be excessively large, at least to ensure that t_n is greater than or equal to 3ms. When t_n is less than 10ms, it should be controlled to be approximately a multiple of 3ms to maintain the integrity of the electrical signal. Consequently, the value of n is not a fixed constant, but varies with the flapping frequency.

The mathematical model of flapping-wing motion angular displacement is established:

$$\theta(t) = \frac{1}{2}Ag(f, t) + \theta_0 \quad (2)$$

where A represents the absolute value of the difference between the upper limit angle and the lower limit angle of a flapping-wing stroke. $g(f, t)$ is the periodic function that characterizes the gait of flapping-wing motion, and θ_0 is the preset initial angular displacement phase. Given the unique nature of the micro servoactuator driven FWAV, $g(f, t)$ can take various forms, such as sine wave, triangle wave, square wave, or any other functional shape. By adjusting $g(f, t)$, both angular displacement and angular velocity can be controlled simultaneously.

Through pulse width modulation, the electrical signal

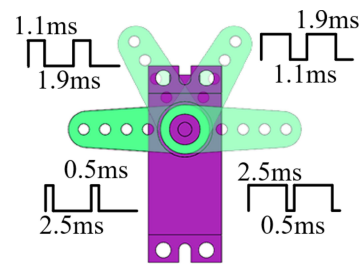


Fig. 1 Pulse width modulation principle for micro servoactuator

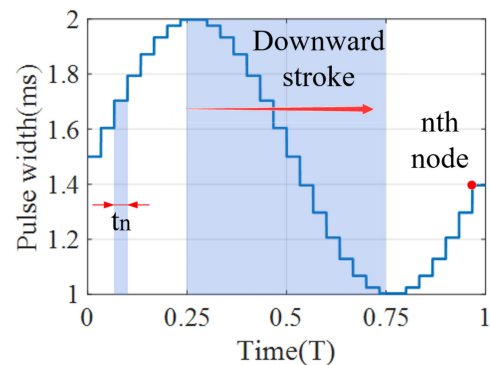


Fig. 2 Generating principle of flapping-wing motion CSSF

driving equations of the left and right micro servoactuators are set as follows:

$$t_{hL} = \sigma\theta(t) + 1520\mu s + e_L \quad (3)$$

$$t_{hR} = t_n - t_{hL} + e_R \quad (4)$$

where the subscripts L and R denote the left and right sides, respectively. t_h represents the high-level portion of each pulse width length, σ signifies the resolution between angular displacement and pulse width, and e denotes the error term. Given the mirror symmetry in the wing motion laws on both sides, the actual angular displacement is opposite, necessitating control of both sides' wings through a complementary pulse width modulation method. Furthermore, accounting for mechanical assembly errors, an error term is introduced to align the actual initial angular displacement phase with θ_0 in the mathematical model. The desired flapping motion can be generated by analyzing the remote control signals received by the micro servo actuator through Eqs. (1)-(4).

3 Motions and Dynamics Parameterization

Aerodynamics research has always been an integral component of the field of FWAV. However, in traditional flapping wing mechanism research, where the wing's motion attitude remains fixed, few studies consider both motion and

dynamic parameters. Most aerodynamic modeling research is conducted at specific flow velocities, flapping frequencies and amplitudes. Nevertheless, the flexible wings of bird-like FWAV result in varying flow states due to different flapping frequencies and amplitudes, leading to significant differences in aerodynamic characteristics [26]. Therefore, leveraging the high controllability of bird-like FWAV driven by micro servoactuator, this paper integrates motion and dynamic parameters for comprehensive research and discussion.

3.1 ALM of Flapping-Wing Motion

Existing research [13, 27, 28] generally posits that the thrust generated by flapping-wing motion might be proportional to the square of the flapping frequency. However, this conclusion isn't entirely confirmed in the scale of bird-like FWAV. In the low-frequency flapping phase, the thrust produced by bird-like FWAV approximately adheres to a quadratic growth relationship. Nonetheless, as the flapping frequency escalates, its thrust generation capacity asymptotically saturates [29]. This phenomenon is more conspicuous in the case of micro servoactuator driven FWAV. With a continued increase in the control input's flapping frequency, the thrust it generates may even decrease instead of rising. This arises from the fact that the micro servoactuator driven flapping-wing motion exhibits noticeable amplitude attenuation [23].

The limitation on motor speed primarily impacts the upper limit of the flapping frequency in traditional flapping-wing motion generation mechanisms because flapping amplitude remains constant. However, there exists an antagonistic relationship between flapping amplitude and frequency in the case of micro servoactuator driven FWAV. As the flapping wing's frequency continuously increases, the amplitude of the flapping-wing motion experiences a corresponding loss. Therefore, it is imperative to investigate the rational configuration of these two parameters. The efficacy of this configuration scheme can be assessed by measuring the relationship between active thrust and power consumption.

The mathematical model for angular velocity during each flapping stroke is:

$$\omega_\lambda = \frac{A}{2} \frac{\partial}{\partial t} g(f, t) \quad (5)$$

For the non-second-order differentiable periodic function, a third-order Fourier fitting function can be used as a substitute in the calculations. The maximum angular speed limit of micro servoactuator at various flapping frequencies is:

$$\omega_s = k_f k_T \omega_m \quad (6)$$

where ω_m represents the no-load limit speed of micro ser-

voactuator under the test voltage, k_T is the speed loss coefficient resulting from the external load, and k_f is the speed loss coefficient attributed to the frequent positive and negative rotations of the flapping-wing motion law.

Consequently, the limit amplitude at any flapping frequency can be calculated:

$$A_m = \frac{2\omega_s}{\left\| \frac{\partial}{\partial t} g(f, t) \right\|_\infty} \quad (7)$$

3.2 FWDM Based on Quasi-Steady (QS) Model

The dynamic model of flapping-wing motion primarily comprises the aerodynamic and inertial force components. Previous studies have rarely discussed the proportion of inertial forces, primarily due to the constraints posed by the fixed parameters of flapping-wing motion and the intricate analytical complexities associated with fluid-solid coupling problems involving flexible deformations. During the flapping-wing motion strokes, inertial force exhibits equal-amplitude oscillations and do not yield average net force in any direction. But in fact, because the moment when the rotating circulation component in the air resistance reaches its peak differs from the moment when the inertial force attains its peak by a quarter of the flapping period, inertial force may lead to advanced chordwise torsion on the airfoil. This will trigger more asymmetric oscillation components and result in higher effective thrust. In this section, the FWDM is established to theoretically investigate the influence of the inertia force ratio on the flexible deformation of the airfoil.

Given the highly time-varying nature of flapping-wing motion, the traditional steady model is no longer applicable. Instead, the QS model serves as a suitable approach for describing the instantaneous aerodynamic forces in flapping-wing motion, making it the most employed modeling method.

The establishment of the QS model is to discretize the flapping-wing strokes along the time axis [30], and the continuous wing surface change is regarded as a series of stationary phases. In each steady state, the blade element method is applied to analyze the aerodynamic characteristics, and the aerodynamic force at each discrete time point is determined by integrating across the spanwise direction. Ultimately, this process yields a time-varying aerodynamic curve composed of densely spaced discrete points.

The instantaneous dynamic equation at each discrete time point is established:

$$F_{inst} = F_{tran} + F_{rot} + F_m \quad (8)$$

where F_{tran} represents the Translational Circulation Component Force (TCCF), F_{rot} represents the Rotational Cir-

ulation Component Force (RCCF), and F_m represents the Inertial Component Force (ICF). In some dynamic modeling studies, special aerodynamic mechanisms such as clap-and-fling effects [8] and wake capture effects [31] are considered. The former has been proven to be a lift-enhancing mechanism present when the wing's amplitude is close to 180 degrees. However, the wing amplitude of bird-like FWAVs is typically within 90 degrees, making the influence of this effect negligible. The latter effect mainly occurs during hovering or low-speed translational flight, but the primary flight mode of bird-like FWAVs is high-speed outdoor movement. Therefore, the influence of these special aerodynamic mechanisms is minimal and has been disregarded in this modeling work. The calculation of the circulation component is based on the widely adopted Dickinson model [31]. The TCCF on any blade element strip can be expressed as:

$$dF_{tran} = \frac{1}{2} \rho v_\infty^2 [C_{L,tran}^2 + C_{D,tran}^2]^{\frac{1}{2}} c_r dr \quad (9)$$

where ρ is the density of the fluid medium, v_∞ is the free flow velocity, c_r is the chord length of the strip, r is the spanwise coordinate of the strip, dr is the width of the strip, $C_{L,tran}$ and $C_{D,tran}$ are the lift coefficient and drag coefficient of the TCCF. These coefficients are solely dependent on the absolute angle of attack α_0 and are determined by the empirical formula:

$$C_{L,tran} = 0.225 + 1.58 \sin(20.13\alpha_0 - 70.2) \quad (10)$$

$$C_{D,tran} = 10.92 - 1.55 \cos(20.04\alpha_0 - 90.82) \quad (11)$$

The RCCF is induced by the passive chordwise torsion of the wing membrane. Particularly when the wing membrane has a large area and high flexibility, the effect of the RCCF cannot be overlooked [32]. The RCCF on any blade element strip can be expressed as:

$$dF_{rot} = \rho \omega_\lambda \omega_\gamma r^2 C_{rot} c_r dr \quad (12)$$

with

$$\omega_\gamma = \frac{\partial}{\partial t} \overline{\gamma}_r \quad (13)$$

$$\overline{\gamma}_r = \frac{\gamma_m}{2} = \frac{dF_N c_r^2}{12EI dr} \quad (14)$$

$$C_{rot} = \pi (0.75 - \hat{x}_0) \quad (15)$$

where ω_γ represents the angular velocity of passive chordwise torsion, $\overline{\gamma}_r$ represents the average chordwise torsion angle of the strip. For simplicity, in the blade element

model, $\overline{\gamma}_r$ can be considered as half of the strip's end section torsion angle γ_m , which can be solved through the end section torsion angle formula under uniform load. dF_N is the instantaneous normal force of the strip on the plane of the flapping shaft, EI is the bending stiffness of the wing in the chord direction. C_{rot} represents the rotating circulation coefficient, $\hat{x}_0$ represents the dimensionless position of the pitch axis relative to the leading-edge axis. For smaller bird-like FWAVs, the leading-edge axis is typically regarded as the flapping axis, hence satisfying $C_{rot} = 0.75\pi$ in this study.

The ICF primarily consists of two parts. One component originates from the wing's own mass during flapping-wing motion, while the other component is generated by the airflow attached to the wing surface. The ICF can be expressed as:

$$dF_m = \frac{\partial}{\partial t} \omega_\lambda (dI_w + dI_a) \quad (16)$$

where dI_w and dI_a are the rotational inertia of the wing membrane and air relative to the rotational axis at the strip, respectively, which can be expressed as:

$$dI_w = \rho_w B c_r r^2 dr \quad (17)$$

$$dI_a = \rho \pi \left(\frac{c_r}{2} \right)^2 r^2 dr \quad (18)$$

where ρ_w is the density of the wing membrane material, and B is the thickness of the wing membrane.

It's necessary to note that in the aforementioned process, dF_{tran} and dF_{rot} act on the blade element strip with chordwise torsion angle, rather than directly on the plane where the flapping shaft is located. To obtain the resultant force along the spanwise integral, the orthogonal decomposition is required in the wing coordinate system of FWAV. In contrast, dF_m doesn't require orthogonal decomposition. Consequently, the expression for the instantaneous normal force of the strip on the plane of the flapping shaft is:

$$dF_N = (dF_{tran} + dF_{rot}) \cos \overline{\gamma}_r + dF_m \quad (19)$$

The instantaneous tangential force is:

$$dF_q = (dF_{tran} + dF_{rot}) \sin \overline{\gamma}_r \quad (20)$$

The process described above is cyclically repeated until the iteration error falls below the predetermined error limit, at which point the cycle is terminated, and the solution result is obtained. To ensure the convergence of the iteration, especially given the significant flexibility of the wing material,

the iteration factor k_n for the average torsion angle is introduced to eliminate the oscillation phenomenon in the iteration. The actual average torsion angle in the $n + 1$ step of the iteration is:

$$\overline{\gamma_{r,n+1}} = k_n \overline{\gamma_{r,n}} + (1 - k_n) \overline{\gamma_{r,n+1}} \quad (21)$$

In the body coordinate system of FWAV, the normal force on the wing surface can be decomposed into orthogonal lift and thrust components:

$$dF_L = dF_N \cos \alpha_0 + dF_q \sin \alpha_0 \quad (22)$$

$$dF_T = dF_N \sin \alpha_0 + dF_q \cos \alpha_0 \quad (23)$$

Assuming the wingspan length is a , various forms of force components can be obtained by integrating along the spanwise direction:

$$F_i = \int_0^a dF_i \quad (24)$$

3.3 PLM of Flapping-Wing Motion

The power load is a valuable metric for assessing the energy utilization efficiency of a certain motion. The presence of significant oscillations in aerodynamic forces during flapping-wing motion implies that the instantaneous aerodynamic force cannot be directly used to calculate the power load. In addition, the average lift is considered to be an indirect force, which is also not suitable for calculating the power load. In fact, under the condition of small angle of attack, the flapping-wing motion cannot directly generate high lift by flapping the wings. Instead, it predominantly relies on the translational circulation mechanism to create lift. However, the free flow velocity term in Eq. (9) cannot be arbitrarily specified. On the contrary, the dominant force in the absence of cruising speed is considered to be the active thrust of flapping-wing motion, primarily caused by the rotating circulation mechanism. As evident from Eq. (12), this force is mainly related to the wing's flapping motion parameters and its flexible deformation factors. Consequently, for the purpose of evaluating the performance of the micro servoactuator driven FWAV studied in this paper, the power load χ is defined as the active thrust generated per unit power consumed by flapping-wing motion without cruising speed:

$$\chi = \frac{\overline{F_T}(v_\infty = 0)}{P_{in}} \quad (25)$$

where P_{in} is the total input power, which can be expressed as:

$$P_{in} = P_0 + \frac{P_w}{\eta} \quad (26)$$

where P_0 represents the no-load power of micro servoactuator generated by frequent forward and reverse rotations during flapping-wing motion, which can be measured through a straightforward experimental approach. η is the mechanical efficiency, typically falls within the range of 90–95%. P_w is the theoretical power demand for wing motion and can be expressed as:

$$P_w = M_w \omega_\lambda \quad (27)$$

where M_w represents the instantaneous moment of the wing, which can be expressed as:

$$M_w = \int_0^a r dF_N \quad (28)$$

The higher the power load value, the more efficient the generation of the same active thrust, which is advantageous for extending the endurance of FWAV.

4 Experimental Validation and Results Discussion

4.1 Experimental Setup

To further investigate the characteristics of micro servoactuator driven FWAV and validate the theory proposed in this study, a series of experiments were conducted using the experimental platform depicted in Fig. 3, mainly included aerodynamic testing and gait capture experiment using a high-speed camera. The flapping-wing motion generating device was connected to a six-axis force/torque sensor (FC6D72C, ForceChina, Inc.), which was securely mounted on an aluminum profile support frame. The FC6D72C sensor had a sampling frequency of 1000 Hz and offered a force resolution of 1/20 N, making it well-suited for the experimental requirements. A fifth-order low-pass Butterworth filter was applied to filter the force/torque data, with the cut-off frequency set to three times the flapping frequency. This allowed the filter to eliminate high-frequency vibrations in the mechanism while retaining reliable force/torque data.

The experimental wing configuration is presented in Fig. 4. The unilateral spanwise length is 300 mm, and the chordwise length is 180 mm. The leading edge of the wing is a 2 mm-diameter carbon fiber rod, and the wing veins are composed of 1 mm-diameter carbon fiber rods. The wing membrane is constructed from P31N reinforced nylon cloth with a thickness of 0.1 mm.

The end of the steering gear arm is marked as 'Mark1',

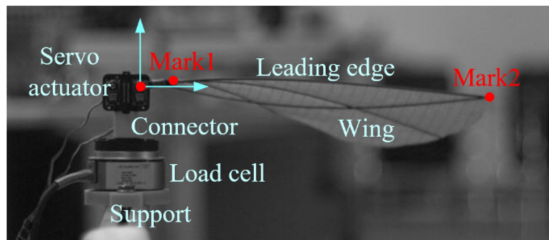


Fig. 3 Experimental platform and flapping-wing motion generating device

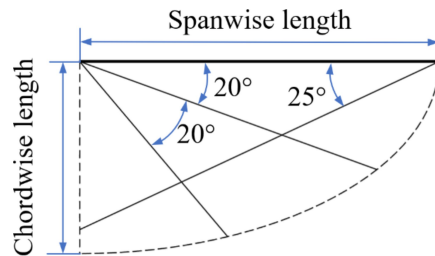


Fig. 4 Wing configuration

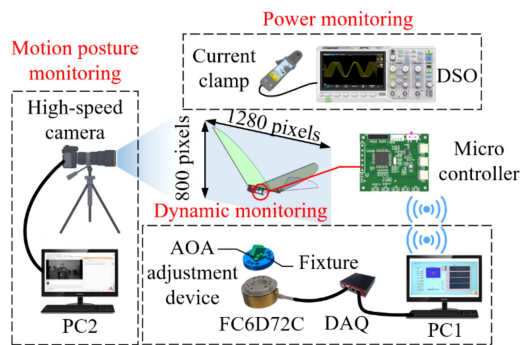


Fig. 5 Test system schematic diagram

and the wing tip is marked as ‘Mark2’. The pixel coordinates of these two marker points in a two-dimensional coordinate system, with the steering gear axis as the origin, are captured by a high-speed camera (VEO-E310L, Vision Research, Inc.) at a frequency of 1000 Hz. This allows for tracking and testing the attitude information of the flapping-wing motion. The power consumed during flapping-wing motion is measured using a current clamp and Digital Storage Oscilloscope (DSO), with a sampling rate of 500 kSa/s. The motion control and signal acquisition principles for the whole system are illustrated in Fig. 5.

4.2 Amplitude Loss Characteristics

The amplitude loss phenomenon in servo-driven flapping-wing motion is essentially due to the speed decay of the servo, caused by two main factors: (1) the unavoidable acceleration and deceleration phases during frequent direction changes; and (2) fluctuating loads caused by aerodynamic

damping. The former is inevitable even in an unloaded (without wing) operating state because, as the flapping frequency increases, the time length of each acceleration and deceleration phase remains constant, but its proportion of the total cycle time increases. The latter generates a reactive torque that reduces the angular acceleration output by the servo, thereby extending the time length of the acceleration and deceleration phases.

To quantify these effects, the average speed (total stroke/total time) over several flapping cycles was measured at different flapping frequencies and CSSFs. Figure 6(a) and Fig. 6 (b) present the experimental results. Figure 6 (a) shows the results under unloaded conditions (denoted as w.o.L.), where the flapping frequency was gradually increased from 2 Hz to 6 Hz. The CSSFs tested included typical sine (denoted as Si.), triangle (denoted as Tr.), and square waves (denoted as Sq.) [25]. It can be observed that under unloaded conditions, the sine wave CSSF exhibits the fastest decay trend with increasing frequency due to the longer time required for acceleration and deceleration, followed by the triangle wave, and then the square wave. Figure 6 (b) shows the data under normal flapping-wing motion conditions (denoted as w.L.). It is evident that the square wave CSSF results in the least speed loss at low frequencies, implying greater amplitude at the same frequency, which is consistent with the conclusions of Huang et al. [25]. However, beyond a certain critical frequency, the speed loss for the square wave CSSF increases sharply due to the large instantaneous damping force caused by RCCF, leading to servo stalling. The abrupt changes in angular displacement of the square wave signal result in instantaneous flapping angular velocities much higher than those of the other two signals. According to Eq. (12), the corresponding RCCF increases, hindering the normal operation of the servo. The triangle wave CSSF occasionally exhibits this phenomenon at flapping frequencies reaching 6 Hz, whereas the sine wave CSSF remains stable below 8 Hz.

Figure 6 (c) shows the time history at a frequency of 4 Hz from Fig. 20(b), illustrating the actual wing motion (solid line) tracking the three forms of CSSF (dotted lines) under the same Preset Amplitude (PA). The fit between the curves is measured by calculating the R^2 coefficient, with values of 0.865, 0.937, and -0.553 for the sine, triangle, and square waves, respectively. This indicates that the tracking is best when using the triangle wave control signal, whereas the square wave control signal results in almost complete loss of control. The corresponding dynamic data were tested, as shown in Fig. 6 (d). The force measurement data exhibit normal behavior for the sine and triangle wave signals, whereas the square wave signal shows significantly reduced force measurements and inconsistent periodicity. The mean thrust values for the three signals are 0.165 N, 0.170 N, and 0.096 N, respectively, confirming that with the same flap-

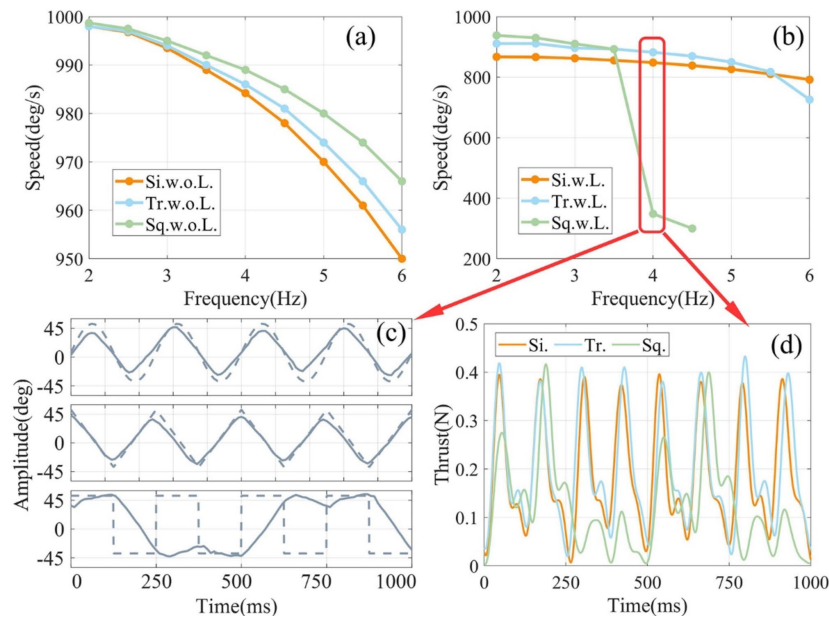


Fig. 6 Impact of different motion parameters on amplitude loss and aerodynamic characteristics. **a** Speed loss under unloaded conditions; **b** Speed loss during normal flapping-wing operation; **c** Time history of wing angle at 4 Hz; **d** Corresponding force measurement data

ping frequency (under normal operating conditions), the smaller the speed loss, the better the thrust performance. In summary, it is concluded that triangle and sine wave CSSF can produce lower amplitude losses within the frequency range required for flapping flight. They have moderate response angular velocities, which accelerate the acceleration and deceleration process while keeping the reactive torque from RCCF below the stall torque. Considering that the thrust performance of the triangle wave is only about 3% higher than that of the sine wave, and the sine wave has a larger reliable operating frequency range, it is concluded that the sine wave CSSF is the most suitable for flapping-wing motion. All subsequent research will be based on the sine wave CSSF.

Based on the experimentally determined maximum rotational speeds at different frequencies, corresponding amplitude parameters were designed. This approach aims to prevent the actual wing motion from deviating too much from the preset CSSF, thereby avoiding abnormal operating conditions. Maximum amplitude tests were conducted at various flapping frequencies, and the amplitude response characteristics are shown in polar coordinates in Fig. 7. At each flapping frequency, the PA was continuously increased. When the actual amplitude was less than 90% of the PA, it was considered that the servo's maximum speed was very close to the limit for that flapping frequency. The actual amplitude at this point was recorded as the maximum amplitude for that frequency. The summarized data is shown in Fig. 8, where the experimental data (denoted as Exp. Data) shows minimal error compared to the model data calculated

using Eq. (7) (denoted as Mod. Data), validating the accuracy of the ALM.

This adaptive PA, designed according to the ALM, can result in better aerodynamic characteristics for the flapping-wing motion. Force measurement experiments considering the ALM were conducted to validate this, and the experimental data is shown in Fig. 9. It is evident that when the PA is set to a fixed value, the aerodynamic characteristics perform well only within a very narrow flapping frequency range. This is because, at lower flapping frequencies, the servo has not reached its maximum operating power; at higher flapping frequencies, the actual wing response trajectory deviates from the preset CSSF, resulting in higher speed loss. However, the adaptive PA ensures good aerodynamic performance at any frequency, making it an effective improvement method for enhancing propulsion performance in micro servoactuator driven FWAVs. All subsequent modeling work has adopted this adaptive PA.

4.3 Leading Chordwise Torsion Effect

In the case of traditional flapping-wing motion generation mechanisms, where the motion attitude remains fixed, a higher flapping frequency generally results in greater thrust. However, once a certain frequency threshold is surpassed, the aerodynamic enhancements become less pronounced, and aerodynamic efficiency starts to decline [33]. Lang et al. [27] found that there exists an optimal flapping frequency for each airfoil to achieve maximum efficiency. This principle holds true for the micro servoactuator driven FWAVs. How-

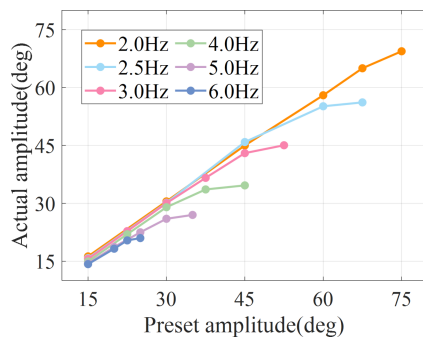


Fig. 7 Amplitude response characteristics at different frequencies

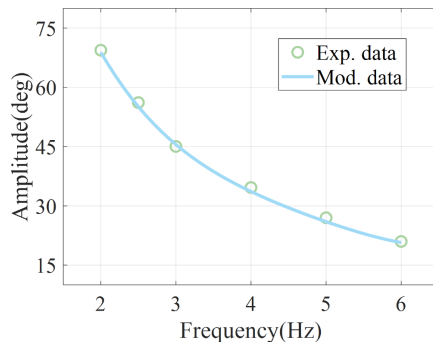


Fig. 8 Limit amplitude at different frequencies

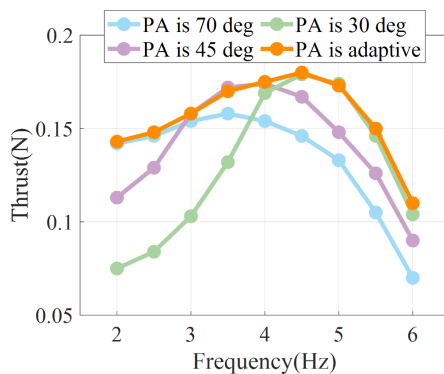


Fig. 9 The aerodynamic characteristics under different PA settings

ever, due to the inherent antagonistic relationship between flapping amplitude and frequency, the impact of different distribution ratios of motion parameters on aerodynamic characteristics becomes more intricate and multifaceted.

Superficially, as the flapping frequency increases, the more pronounced amplitude attenuation phenomenon might suggest a monotonous decrease in flapping flight efficiency. However, in reality, this is not the case. The decrease in the limit speed of the micro servoactuator is accompanied by a rapid increase in angular acceleration. Figure 10 displays the angular velocity and angular acceleration of the flapping wing, calculated from the data presented in Fig. 8. As discussed in Sect. 4.2, the RCCF is directly associated with the angular velocity of the flapping-wing motion, while the

ICF is associated with angular acceleration. With the rise in flapping frequency, the proportion of the former component decreases, while the latter component increases. Therefore, the investigation into how to distribute various motion parameters to maximize benefits is essentially to study the influence of the proportion of ICF on the aerodynamic characteristics.

Consequently, two dynamic models have been established. Model_1 considers the influence of ICF, while Model_2 neglects the ICF. The ICF cannot directly generate asymmetric force but enhance the tangential force component of the aerodynamic force on the plane where the flapping shaft is situated by affecting real-time flexible wing surface deformation. Figure 11 illustrates this indirect influence relationship. The peaks of lift and thrust in Model_2 are observed at the highest flapping speed during the downstroke, whereas in Model_1, the peaks shift forward along the timeline. When the flapping-wing motion follows an approximate triangular wave to the transition between the upstroke and the downstroke, the angular velocity direction changes, and instantaneous angular velocity approaches zero. Consequently, in Model_2, lift and thrust are also reduced to zero at this point. However, in Model_1, which considers the influence of ICF, this moment is also the moment of maximum acceleration. As a result, the ICF reaches its peak earlier, causing the chordwise torsion of the wing to invert prematurely, resulting in the additional thrust component depicted in the red-shadowed section of Fig. 11.

Figure 12 depicts this phenomenon from the time-varying trend of flexible deformation of the wing surface. It is evident that, as the downstroke is about to commence, the chordwise torsion angle in Model_1 turns positive in advance. Similarly, as the upstroke is about to commence, the chordwise torsion angle in Model_1 turns negative in advance. The leading chordwise torsion effect induced by the inertial force effectively enhances the thrust component generated by the flapping-wing motion. Figure 13 shows the gain variation trend of Model_1 relative to Model_2 at different flapping frequencies. It can be seen that this gain peaks at 0.086 N around a flapping frequency of 4.4 Hz. The thrust gain generated by the leading torsional effect is inherent in flapping wing motion, and the FWDM (Model_1) including ICF can accurately reflect the influence of this phenomenon on aerodynamic characteristics.

4.4 Model Application and Error Analysis

The modeling method proposed in this study can effectively analyze the impact of amplitude decay and advanced twisting on aerodynamic characteristics. Therefore, it offers high predictive accuracy for servo-driven FWAVs and can be utilized for optimizing flapping motion parameters.

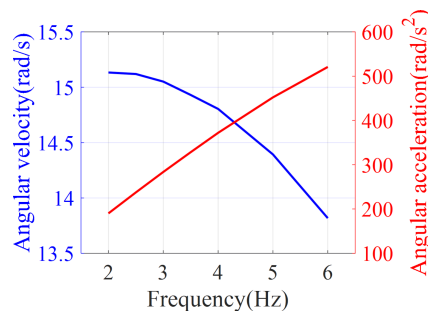


Fig. 10 The variation trend of dynamic parameters with flapping frequency

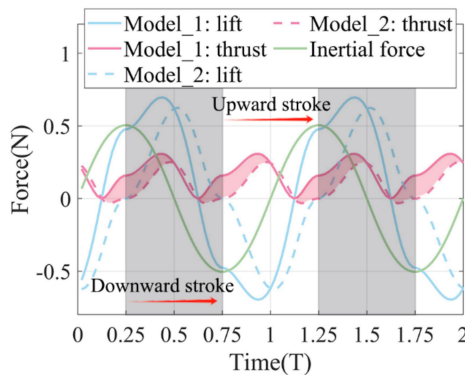


Fig. 11 The gain of ICF on flapping-wing thrust performance

Based on the conclusions drawn in Sects. 4.2 and 4.3, for servo-driven FWAVs, increasing the flapping frequency is associated with a decrease in the maximum rotational speed and an increase in the contribution of inertial force components. The former reduces aerodynamic efficiency, while the latter enhances it. To find a balance that maximizes benefits, the dynamic modeling method was employed to explore the trends in thrust, power, and power load at different flapping frequencies, and these were compared with experimental data, as shown in Figs. 14, 15 and 16.

It can be seen that as the flapping frequency gradually increases, the thrust initially increases and then decreases, the power continuously increases, and the power load remains stable before rapidly decaying. The model's predicted results align with the experimental results. Calculating the Absolute Percentage Error (APE) reveals that within the commonly encountered flapping frequency range, the proposed dynamic model maintains an estimation error of less than 10% for thrust performance, under 5% for power estimation, and below 13% for power load estimation.

According to the model's calculations, the designed FWAV prototype can achieve its maximum thrust performance at a frequency of 4.4 Hz with a PA of 30 degrees, reaching a thrust value close to 0.18 N. Additionally, the power load can maintain a relatively high level (around 0.05 N/W) when the flapping frequency is below 5 Hz with

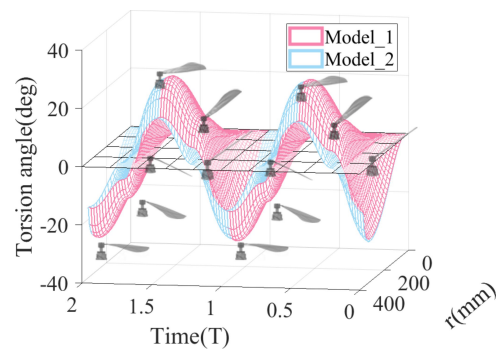


Fig. 12 Influence of ICF on flexible deformation

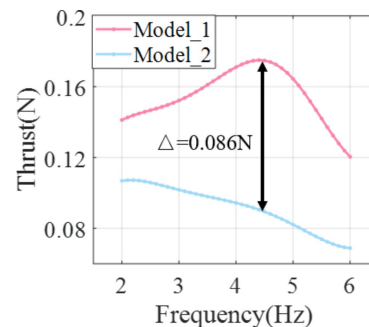


Fig. 13 Thrust gain generated by the advanced twisting effect caused by inertial force

the corresponding adaptive PA. This indicates that in practical applications, the suitable motion parameters derived from the proposed dynamic model can enhance the payload capacity of flapping-wing vehicles, while also extending flight time and improving flight efficiency.

5 Conclusion

This study addresses the specificity of servo-driven FWAVs by proposing corresponding ALM, FWDM, and PLM. It quantifies the relationship between motion parameters and aerodynamic characteristics and achieves comprehensive validation through motion capture experiments, power monitoring experiments, and force monitoring experiments. The experimental results indicate that the amplitude decay phenomenon generated by sine wave CSSF is relatively stable and more suitable for flapping motion. Combined with the adaptive PA, this approach can effectively enhance the thrust performance of flapping motion. The proposed FWDM considers the effects of amplitude decay characteristics and advanced twisting effects, with prediction APEs for thrust, power, and power load indicators being less than 10%, 5%, and 13%, respectively. This high-precision model can effectively optimize motion parameters, allowing for better performance in flapping-wing motion. In the future, we will continue to explore the application of the proposed dynamic model in further optimizing structural

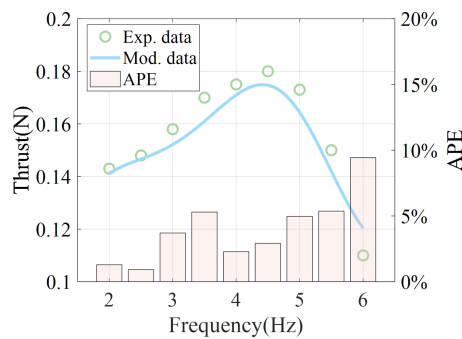


Fig. 14 The variation trend of thrust force with flapping frequency

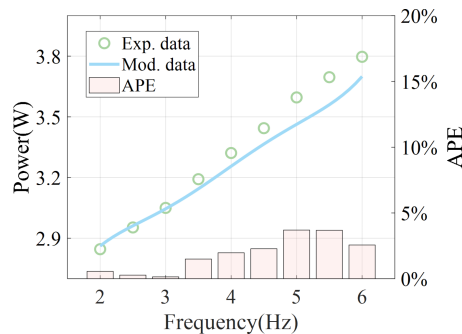


Fig. 15 The variation trend of power with flapping frequency

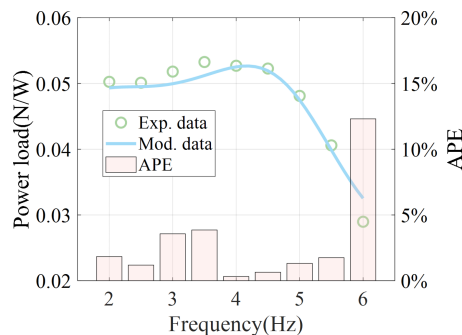


Fig. 16 The variation trend of power load with flapping frequency

design and guiding controller design to meet task requirements in adverse weather conditions or complex obstacle environments.

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Data Availability Data will be made available on request.

Declarations

Conflict of interest The authors declared that they have no conflicts of interest.

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